

Sarah, 35 · 4 months of weight loss, heat intolerance, palpitations, anxiety · **diffuse non-tender goiter** · **exophthalmos** · TSH undetectable, free T4 high · TSI antibody positive · **Graves disease**

BIO 004 · WEEK 6 · CASE DEEP DIVE

Endocrine Anatomy

A 35-year-old whose body is running too hot, a gland in her neck the size of a plum, and her own antibodies driving it.

PATIENT CASE

Sarah is 35. Over four months she has **lost 15 pounds without trying**, eats more than before, sweats easily, can't tolerate warm rooms, and has had palpitations and tremors. She's anxious, sleeping poorly. Her family says her eyes "look different."

On exam HR 110, BP 140/70, fine resting tremor, warm moist skin, **diffusely enlarged non-tender thyroid (goiter)**, **proptosis** of both eyes. Labs: **TSH undetectable, free T4 markedly elevated, total T3 elevated**. Thyroid-stimulating immunoglobulin (TSI) is positive. The diagnosis is **Graves disease** — an autoimmune disorder in which antibodies activate the TSH receptor, driving overproduction of thyroid hormone.

The endocrine system is a constellation of glands scattered through the body, each producing hormones that travel through the bloodstream to act on distant targets. Sarah's case rests on the anatomy of one gland and the master controller that normally throttles it.

GUIDING QUESTION

Where are the major endocrine glands, what does each produce, and what happens when an autoimmune antibody mimics the master signal that should be tightly regulated?

1 PLAN

Glands scattered, hormones in blood, targets distant

Unlike the nervous system (fast, point-to-point), the endocrine system is slow and broadcast. A gland releases a hormone into the blood; the hormone reaches every cell but only cells with the right receptor respond. Two big types of hormones: **steroid hormones** (lipid-soluble, diffuse through membranes, bind intracellular receptors) and **peptide hormones** (water-soluble, bind surface receptors, trigger second-messenger cascades).

2 HYPOTHALAMUS-PITUITARY

The master controller and its two-part executive

The **hypothalamus** (floor of the diencephalon) controls the pituitary through two pathways:

● **Anterior pituitary (adenohypophysis)**
Glandular tissue. Receives *releasing hormones* from the hypothalamus via a portal venous system. Produces: TSH, ACTH, FSH, LH, GH, prolactin. *TSH is what Sarah's antibody mimics.*

● **Posterior pituitary (neurohypophysis)**
Neural tissue, an extension of the hypothalamus. Releases *ADH* (vasopressin) and *oxytocin*, which are made in the hypothalamus and transported down the axons.

The whole gland sits in the **sella turcica** of the sphenoid bone. Pituitary tumors can compress the optic chiasm just above → bitemporal hemianopsia.

Anterior neck, two organs, four hormones

The **thyroid** sits anterior to the trachea, just below the larynx, with two lobes connected by an isthmus. Functional unit: the *thyroid follicle* — sphere of cuboidal follicular cells around a colloid-filled lumen. Follicular cells synthesize T3 and T4 (iodine-containing hormones) from colloid stores. Parafollicular (C) cells between follicles produce *calcitonin*.

● T3 and T4

Set metabolic rate. Drive nearly all of Sarah's symptoms when in excess.

● Calcitonin

Lowers blood calcium by inhibiting osteoclasts.
Minor role in humans.

The **parathyroid glands** (usually 4, on the posterior thyroid) produce *parathyroid hormone (PTH)*, which raises blood calcium. Critical: surgeons performing thyroidectomy must spare them.

Why Graves' produces what it does. TSI is an autoantibody that binds and activates the TSH receptor on follicular cells. It mimics TSH but is not under negative feedback. The thyroid grows (goiter) and pours out T3/T4. TSH drops to undetectable because the pituitary senses high T3/T4 (normal feedback) — but the antibody keeps stimulating the gland anyway.

A gland with two parts that have nothing in common

The **adrenal glands** sit on top of each kidney. Each is two organs glued together:

● Adrenal cortex (outer)

Three zones, three steroid families. *Zona glomerulosa*: aldosterone (mineralocorticoid; Na⁺/water retention, K⁺ secretion). *Zona fasciculata*: cortisol (glucocorticoid; metabolism, immune suppression, stress response). *Zona reticularis*: androgens.

● Adrenal medulla (inner)

Modified sympathetic ganglion. Chromaffin cells release *epinephrine and norepinephrine* into the bloodstream as the "long-distance" arm of sympathetic outflow.

Mnemonic for cortex zones: GFR — Glomerulosa, Fasciculata, Reticularis. And the hormones: "*The deeper you go, the sweeter it gets*" (salt → sugar → sex).

1% of pancreatic tissue, 100% of glucose control

Embedded in the otherwise-exocrine pancreas are 1-2 million **islets of Langerhans** (covered on the Accessory Digestive Organs page in their exocrine context). Three main cell types:

● Alpha (~20%)

Glucagon. Raises blood glucose between meals.

● Beta (~70%)

Insulin. Lowers blood glucose after meals. Type 1 diabetes destroys these; type 2 has insulin resistance + relative insufficiency.

● Delta (~5%)

Somatostatin. Inhibits both insulin and glucagon.
Local modulator.

Three more endocrine players

Gonads: ovaries (estrogen, progesterone) and testes (testosterone). Covered on the reproductive pages. **Pineal gland:** small endocrine gland in the epithalamus. Produces *melatonin*, the sleep/circadian hormone. **Thymus:** where T cells mature (covered on the Lymphatic System page). Also has endocrine functions through thymic hormones (thymosin, thymopoietin) involved in T-cell development.



DRAW ZONE 1 -- Sketch the body and label every endocrine gland: hypothalamus, pituitary, pineal, thyroid, parathyroids, thymus, adrenals, pancreatic islets, ovaries/testes.

Use a body outline with arrows from each gland. Star the ones Sarah's antibody attacks (thyroid) and the master controller (hypothalamus-pituitary).



DRAW ZONE 2 -- Diagram the hypothalamus-pituitary-thyroid axis. Show TRH → TSH → T3/T4. Then add Sarah's TSI antibody bypassing the normal feedback at the receptor.

Show how undetectable TSH and high T4 coexist — TSI keeps the gland firing even when pituitary tries to shut it down.

RETURNING TO SARAH

An antibody that won't take no for an answer

Every one of Sarah's symptoms is the metabolic effect of too much T3/T4 acting on tissues that are working normally — they're just being pushed too hard.

"Weight loss while eating more"

Elevated T3/T4 raises basal metabolic rate. Calories are burned faster than she can eat them.

"Heat intolerance, warm moist skin"

Thyroid hormone increases thermogenesis. Body dumps heat through skin vasodilation and sweating.

"Palpitations, HR 110"

T3 upregulates beta-adrenergic receptors on the heart. Hyperthyroid patients are effectively in a hyper-sympathetic state.

"Diffuse non-tender goiter"

TSI uniformly stimulates the whole gland → diffuse enlargement. Contrast with nodular goiter (focal) or tender thyroiditis (inflammation).

"Exophthalmos"

TSI also targets fibroblasts in the orbit, triggering glycosaminoglycan accumulation behind the eyes. Specific to Graves disease (not other causes of hyperthyroidism).

"Why TSH undetectable and T4 high together"

TSI is driving thyroid output independent of TSH. The pituitary senses the high T4 and shuts down TSH (normal feedback). But the gland keeps producing because TSI is acting on the receptor directly.

Hypothalamus, pituitary, thyroid, receptor. Sarah's body is being run by an antibody that mimics a signal that should be carefully regulated.

